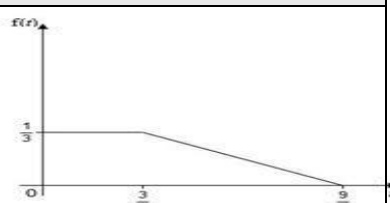


Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Obtains w from $\int_0^{15} w l \, dl = 1$ Allow one error if method correct	AO1.1a	M1	$\int_0^{15} w l \, dl = w \left[\frac{l^2}{2} \right]_0^{15}$ $= \frac{225w}{2} = 1$
Obtains $E(L^2)$ by integrating $\int l^2 \times f(l) \, dl$ FT 'their' value for w	AO1.1a	M1	$w = \frac{2}{225}$ Hence $E(L^2) = \int_0^{15} l^2 \times \frac{2}{225} l \, dl$
Obtains $E(S)$ by evaluating $\frac{1}{15}(0+1+2+3+4+5)$ Allow one error if method correct	AO1.1a	M1	$= \frac{2}{225} \left[\frac{l^4}{4} \right]_0^{15} = \frac{225}{2}$ $E(S) =$
Uses $E(T) = E(L^2) + \frac{1}{2}E(S)$ (PI)	AO1.1a	M1	$\frac{1}{15}(0 \times 0 + 1 \times 1 + 2 \times 2 + \dots + 5 \times 5)$
Shows that $E(T) = 114 \frac{1}{3}$ Mark awarded if they have a completely correct solution, which is clear, easy to follow and contains no slips AG	AO2.1	R1	$= \frac{55}{15} = \frac{11}{3}$ $E(T) = E(L^2) + \frac{1}{2}E(S)$ $= \frac{225}{2} + \frac{11}{6} = \frac{343}{3} = 114 \frac{1}{3}$ AG
Total 5 marks			

Q2.

Marking Instructions	AO	Marks	Typical Solution
(a) (i) Draws horizontal line $(0, \frac{1}{3})$ to $(\frac{3}{2}, \frac{1}{3})$ and straight line joining	AO1.1a	M1	

	$(\frac{3}{2}, \frac{1}{3})$ to t axis			
	Sketches correct shape, accurate and fully labelled	AO1.1b	A1	
(ii)	Deduces median is $\frac{3}{2}$	AO2.2a	B1	Area under $f(t)$ for $0 \leq t \leq \frac{3}{2}$ is $\frac{1}{2}$ Median of t is $\frac{3}{2}$
(b)	States integral required for $E(T)$	AO1.1a	M1	$E(T) = \int_0^{\frac{3}{2}} \frac{1}{3} t dt + \int_{\frac{3}{2}}^{\frac{9}{2}} \frac{t(9-2t)}{18} dt$
(i)	Obtains $E(T)$ correctly	AO1.1b	A1	$= \left[\frac{1}{6} t^2 \right]_0^{\frac{3}{2}} + \left[\frac{t^2}{4} - \frac{t^3}{27} \right]_{\frac{3}{2}}^{\frac{9}{2}} = \frac{13}{8}$
(ii)	Uses definition for variance (PI)	AO1.1a	M1	$\text{Var}(T) = E(T^2) - (E(T))^2$
	Obtains $\text{Var}(T)$	AO1.1b	A1	$\text{Var}(T) = \frac{15}{4} - \left(\frac{13}{8} \right)^2 = \frac{71}{64}$
	Obtains $\text{Var}(4T - 5)$ (Follow through 'their' $\text{Var}(T)$)	AO1.1b	A1F	$\text{Var}(4T - 5) = 16 \text{Var}(T) = \frac{71}{4}$
Total 8 marks				

Q3.

(a) Putting $\frac{t}{216} = 0.9$

M1

$$t = 5.793$$

5.79 to 5.80

A1

41 days.

Accept 40 days in this context

A1

3

- (b) Attempt to differentiate $F(t)$

ct^2 seen

M1

$$f(t) = \frac{1}{72} t^3 \quad 0 \leq t \leq 6$$

Condone domain missing here

A1

$$= 0 \quad \text{otherwise}$$

For **complete** function

A1

3

- (c) Attempt to integrate $tf(t)$ from 0 to 6

Using their $f(t)$ from **(b)** ct^4 seen

M1

$$E(T) = 4.5$$

A1

Attempt to integrate $t^2f(t)$ from 0 to 6

Using their $f(t)$ from **(b)** ct^5 seen

M1

$$E(T^2) = 21.6$$

A1

$$\text{Var}(T) = E(T^2) - E(T)^2$$

Applied in this case. Dependent on both M1

m1

$$= 21.6 - 4.5^2 = 1.35$$

A1

6

- (d) S.d. = $\sqrt{1.35} = 1.162$

For $\sqrt{\text{their Var}}$ $0 < \text{Var}(T) < 9$

M1

Use of $F(5.662)$

For $F(\text{their s.d.} + \text{their } E(T))$ provided $0 < \text{Total} < 6$

m1

$$1 - \frac{5.662^3}{216}$$

m1

$$= 0.160$$

AWFW 0.159 to 0.161

A1

4

[16]

Q4.

(a) (i) $E(T) = \frac{1}{2} (25 + -5) = 10$
CAO

B1

1

(ii) $Var(T) = \frac{1}{12} (25 - -5)^2 = 75$
CAO

B1

1

$$\frac{2}{15}$$

(b) $P(-2 < T < 2) =$ (OE)

Diagram optional



B1

P(magnitude at least 2 minutes)

$$= 1 - P(-2 < T < 2)$$

$$= 1 - \frac{4}{30}$$

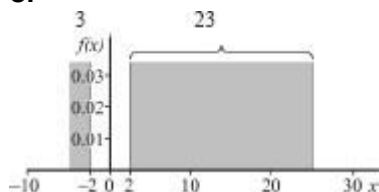
M1

$$= \frac{13}{15} \text{ (OE)} = 0.867$$

CAO (AWRT)

A1

or



$$\frac{1}{30} (3 + 23) = \frac{26}{30} = \frac{13}{15}$$

Alternative

$$P(T > 2) = \frac{23}{30} (0.766\bar{6})$$

or $P(T < -2) = \frac{1}{10}$

B1

or

$$\int_{-2}^0 \frac{1}{30} dt + \int_0^{25} \frac{1}{30} dt = \frac{1}{10} + \frac{23}{30} = \frac{13}{15}$$

$P(\text{magnitude at least 2 minutes})$

$$= P(T < -2) + P(T > 2)$$

$$= \frac{13}{15} \text{ for M1 A1}$$

or

$$1 - \int_{-2}^2 \frac{1}{30} dt = 1 - \left[\frac{t}{30} \right]_{-2}^2$$

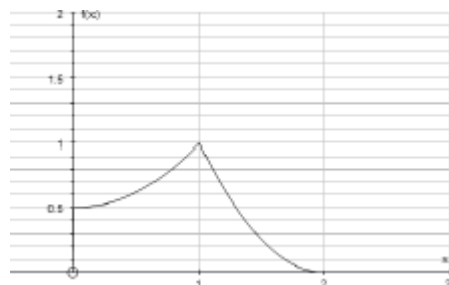
$$= 1 - \frac{4}{30} = \frac{26}{30} = \frac{13}{15}$$

3

[5]

Q5.

(a)



B1 for axes

B1 for curve from (0, 0.5) to (1, 1)

B1 for curve from (1, 1) to (2, 0)

B3

3

(b)
$$P(X \leq 1) = \int_0^1 \frac{1}{2}(x^2 + 1) dx$$

M1

$$= \left[\frac{x^3}{6} + \frac{x}{2} \right]_0^1$$

A1

$$= \left[\frac{1}{6} + \frac{1}{2} \right] = \frac{2}{3}$$

0.667

A1

3

(c)
$$E(X^2) = \int_0^1 x^2 \times \frac{1}{2}(x^2 + 1) dx + \int_1^2 x^2 (x - 2)^2 dx$$

both integrals seen

M1

$$= \left[\frac{x^5}{10} + \frac{x^3}{6} \right]_{x=0}^{x=1} + \left[\frac{x^5}{5} - x^4 + \frac{4x^3}{3} \right]_{x=1}^{x=2}$$

A1A1

$$= \left(\frac{1}{10} + \frac{1}{6} \right) + \left(\left[\frac{32}{5} - 16 + \frac{32}{3} \right] - \left[\frac{1}{5} - 1 + \frac{4}{3} \right] \right)$$

dep(M1)

m1

$$= \frac{4}{5}$$

AG

A1

5

(d) (i)
$$E(X) = \frac{19}{24} \text{ and } k\text{Var}(X) = 499$$

$$\text{Var}(X) = E(X^2) - E^2(X)$$

$$= \frac{4}{5} - \left(\frac{19}{24} \right)^2$$

M1

$$= \frac{499}{2880} \quad (0.173)$$

A1

$$\Rightarrow k = 2880$$

CAO

A1

3

$$\begin{aligned} \text{(ii)} \quad E(5X^2 + 24X - 3) \\ = 5E(X^2) + 24E(X) - 3 \end{aligned}$$

M1

$$\begin{aligned} &= 5 \times \frac{4}{5} + 24 \times \frac{19}{24} - 3 \\ &= 20 \end{aligned}$$

CAO

A1

2

$$\text{(iii)} \quad \text{Var}(12X - 5) = 144\text{Var}(X)$$

M1

$$\begin{aligned} &= 144 \times \frac{499}{2880} \\ &= \frac{499}{20} \quad \text{or } (24.95) \end{aligned}$$

CAO (AWFW 24.9 to 25)

A1

2

[18]

Q6.

$$\begin{aligned} \text{(a)} \quad &P\left(-\frac{3c}{4} < X < \frac{3c}{4}\right) \\ &= \frac{\frac{3c}{4} + c}{4c} - \frac{-\frac{3c}{4} + c}{4c} \\ &\quad \text{or} = \frac{3c}{2} \times \frac{1}{4c} \end{aligned}$$

M1

$$= \frac{6c}{16c}$$

$$= \frac{3}{8} \text{ or } 0.375$$

CAO

A1

2

(b) For $-c \leq x \leq 3c$

$$f(x) = \frac{d}{dx} \left(\frac{x+c}{4c} \right)$$

use of $f(x) = F'(x)$

M1

$$= \frac{1}{4c}$$

For $x > 3c$ and $x < -c$

$$f(x) = \frac{d}{dx} (F) = 0$$

for $\frac{1}{4c}$ and 0

A1

2

(c) (i) Rectangular distribution:

$$E(X) = \frac{1}{2}(-c + 3c) = c$$

B1

1

(ii) $\text{Var}(X) = \frac{1}{12}(3c - -c)^2 = \frac{4c^2}{3}$

Allow $\frac{16c^2}{12}$

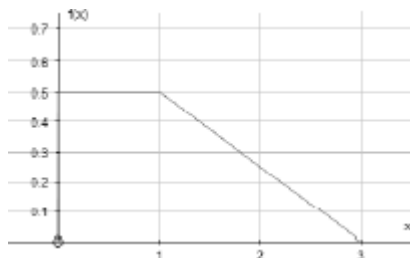
B1

1

[6]

Q7.

(a) Sketch:



1 for straight line $0 \leq x \leq 1$ from $(0, 0.5)$ to $(1, 0.5)$

1 for straight line $1 \leq x \leq 3$ from $(1, 0.5)$ to $(3, 0)$

1 for axes [must have at least $(0, 0.5)$

$(1, 0)$ and $(3, 0)$ labelled]

B3

3

(b) $P(X \leq \eta) = F(\eta) = 0.5$

M1

($\Rightarrow \eta = 1$ (from graph))

AG

A1

2

(c) $\mu = E(X) = \int_0^1 \left(\frac{x}{2}\right) dx + \int_1^3 x \left(\frac{3-x}{4}\right) dx$

Both integrals stated

M1

$$= \left[\frac{x^2}{4} \right]_0^1 + \frac{1}{4} \left[\frac{3x^2}{2} - \frac{x^3}{3} \right]_1^3$$

Either

A1

$$= \frac{1}{4} + \frac{1}{4} \left[\left(\frac{27}{2} - 9 \right) - \left(\frac{3}{2} - \frac{1}{3} \right) \right]$$

Correct limits used on both
integrals + combined dep M1

m1

$$= \frac{1}{4} + \frac{5}{6} \quad (0.25 + 0.83\dot{3})$$

$$= 1\frac{1}{12}$$

(CAO)

A1

(d) Area of Δ

$$= P\left(X > 2\frac{1}{4}\right) = \frac{1}{2} \times \frac{3}{4} \times \frac{3 - 2\frac{1}{4}}{4}$$

M1ft

$$= \frac{3}{32} \times \frac{3}{4} = \frac{9}{128}$$

$$\therefore P\left(X < 2\frac{1}{4}\right) = 1 - \frac{9}{128}$$

M1ft

$$= \frac{119}{128} (0.9296875)$$

A1

Alternative:

For $1 \leq x \leq 3$

$$F(x) = 1 - \frac{1}{8}(3-x)^2$$

M1ft

$$F\left(2\frac{1}{4}\right) = 1 - \frac{1}{8} \times \frac{9}{16}$$

M1ft

$$= \frac{119}{128}$$

CAO

or

$$\int_{2\frac{1}{4}}^3 \frac{3-x}{4} dx \left(= \frac{9}{128} \right)$$

M1ft

$$= 1 \int_{2\frac{1}{4}}^3 \frac{3-x}{4} dx$$

M1ft

$$\begin{aligned}
 &= 1 - \frac{1}{4} \left[3x - \frac{x^2}{2} \right]_{2\frac{1}{4}}^3 \\
 &= 1 - \frac{1}{4} \left[9 - \frac{9}{2} - \frac{27}{4} + \frac{81}{32} \right] \\
 &= 1 - \frac{1}{4} \times \frac{9}{32} = \frac{119}{128}
 \end{aligned}$$

A1

or $(1 - 0.0703125 = 0.9296875)$

Alternative

$$f\left(2\frac{1}{4}\right) = \frac{3}{16} = 0.1875$$

$$P(X < 3\mu - \eta) = P\left(X < 2\frac{1}{4}\right)$$

$$= \frac{1}{2} + \frac{1}{2} \left(\frac{3}{16} + \frac{1}{2} \right) \times 1\frac{1}{4} \quad \text{M1ft}$$

$$= \frac{1}{2} + \frac{55}{128} (0.4296875) \quad \text{M1ft}$$

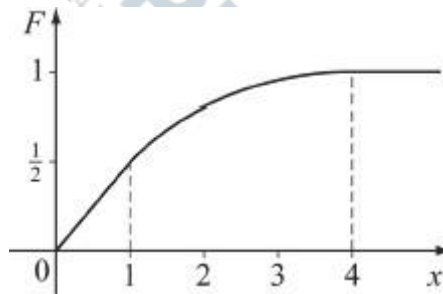
$$= \frac{119}{128} (0.930) \quad \text{A1}$$

3

[12]

Q8.

(a) (i)



B1 for axes 0 to 4 & 0 to 1

B1 for straight line $0 - \left(1, \frac{1}{2}\right)$

B1 for convex curve from $\left(1, \frac{1}{2}\right)$ to $(4, 1)$

B1 for at least the straight line for $x > 4$

B4

4

(ii) $F(q_1) = 0.25 \Rightarrow \frac{1}{2}q_1 = 0.25$

From sketch, or from $F(x)$, Median = 1.

M1

$$\Rightarrow q_1 = \frac{1}{2}$$

$\frac{1}{2}x$ is linear on $(0, 0)$ to $\left(1, \frac{1}{2}\right)$

$$\therefore q_1 = \frac{1}{2}$$

AG

A1

2

(iii) $F(1.6) = 0.744$
 $F(1.7) = 0.775$

M1

$$F(q_3) = 0.75$$

M1

$$\Rightarrow 1.6 < q_3 < 1.7$$

AG

A1

3

(b) (i) $f(x) = F(x)$

M1

$$\Rightarrow f(x) = \frac{1}{2} \text{ for } 0 \leq x \leq 1$$

A1

$$\Rightarrow a = \frac{1}{2}$$

$$f(1) = a = 9\beta$$

B1

$$\int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow$$

$$[ax]_0^1 + \left[\frac{\beta(x-4)^3}{3} \right]_1^4 = 1$$

M1

$$\Rightarrow \text{for } 1 \leq x \leq 4$$

$$f(x) = \frac{1}{54} (3x^2 - 24x + 48)$$

A1

$$= \frac{3}{54} (x^2 - 8x + 16)$$

M1

$$= \frac{1}{18} (x - 4)^2$$

$$\Rightarrow \beta = \frac{1}{18}$$

A1

$$\Rightarrow \alpha + 9\beta = 1 \quad A1$$

Solving: M1

$$\alpha = \frac{1}{2} \text{ and } \beta = \frac{1}{18} \quad A1$$

5

$$(ii) \quad E(X) = \int_0^1 \frac{1}{2} x dx + \int_1^4 \frac{1}{18} (x^3 - 8x^2 + 16x) dx$$

Both seen

M1

$$= \left[\frac{1}{4} x^2 \right]_0^1 + \frac{1}{18} \left[\frac{x^4}{4} - \frac{8x^3}{3} + 8x^2 \right]_1^4$$

A1A1

$$= \frac{1}{4} + \frac{7}{8}$$

Dependent on M1

m1

CAO

A1

$$= 1 \frac{1}{8}$$

5

[19]

Q9.

$$(a) \quad P(X < 0) = F(0)$$

M1

$$= \frac{1}{k+1}$$

A1

2

(b) (i) $f(x) = \frac{d}{dx}(F(x))$
use of

M1

$$= \frac{1}{k+1} \times \frac{d}{dx}(x+1)$$

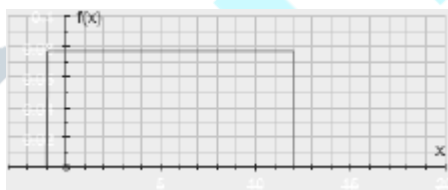
$$= \frac{1}{k+1} \text{ for } -1 \leq x \leq k$$

AG; $\frac{1}{k+1}$ clearly deduced

A1

2

(ii)



B2

2

(c) $\frac{1}{12}(k+1)^2 = \left(\frac{1}{2}(k-1)\right)^2$

M1

$$\frac{1}{12}(k+1)^2 = \frac{1}{4}(k-1)^2$$

$$(k+1)^2 = 3(k-1)^2$$

$$k^2 + 2k + 1 = 3k^2 - 6k + 3$$

M1

$$2k^2 - 8k + 2 = 0$$

$$k^2 - 4k + 1 = 0$$

A1

$$k = \frac{4 \pm \sqrt{16-4}}{2}$$

m1

$$k = 2 - \sqrt{3} \text{ or } k = 2 + \sqrt{3}$$

AG

A1

Alternative:

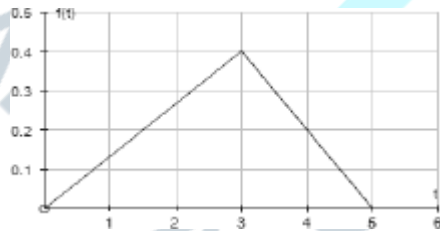
$$\begin{aligned} (k+1)^2 &= (\sqrt{3}(k-1))^2 \\ [(k+1) + \sqrt{3}(k-1)][(k+1) - \sqrt{3}(k-1)] &= 0 \\ k &= \frac{\sqrt{3}-1}{1+\sqrt{3}} \text{ or } \frac{1+\sqrt{3}}{\sqrt{3}-1} \\ k &= 2 - \sqrt{3} \text{ or } k = 2 + \sqrt{3} \end{aligned}$$

5

[11]

Q10.

(a)



line segment on 0 – 3

line segment on 3 – 5

scales (0 – 0.4 vertical; 0 – 5 horizontal)

B1

B1

B1

3

(b) (i) $P(T \leq 2) = \frac{1}{2} \times 2 \times \frac{4}{15}$

M1

$$= \frac{4}{15}$$

(0.267)

A1

2

(ii) $P(2 < T < 4) = 1 - (P(T < 2) + P(T > 4))$

M1

$$= 1 - \left(\frac{4}{15} + \frac{1}{2} \times \frac{1}{5} \right)$$

$$\text{for } P(T > 4) = \frac{1}{10}$$

$$\frac{1}{2} d[(f_2 + f_4) + 2f_3]$$

$$f_2 = \frac{4}{15}; f_4 = \frac{1}{5}; f_3 = \frac{2}{5}; d = 1$$

A1

$$= 1 - \frac{4}{15} - \frac{1}{10}$$

$$= \frac{19}{30}$$

(0.633)

A1

3

(c) $E(T) = \int_0^3 \frac{2}{15} t^2 \, dt = \int_3^5 t \left(1 - \frac{1}{5} t \right) \, dt$

both integrals seen

M1

$$= \left[\frac{2}{45} t^3 \right]_0^3 + \left[\frac{1}{2} t^2 - \frac{1}{15} t^3 \right]_3^5$$

B1B1

$$= \frac{6}{5} + \frac{25}{6} - \frac{27}{10}$$

$$= 2\frac{2}{3}$$

OE

A1

4

[12]

Q11.

(a) $P(X < 0) = F(0)$

M1

$$= \frac{1}{k+1}$$

A1

2

(b) $(q_1 + 1) \times \frac{1}{k+1} = \frac{1}{4}$

alternative (from a sketch)

M1

$$q_1 + 1 = \frac{1}{4}(k+1)$$

A1

$$q_1 = \frac{1}{4}(k+1) - 1$$

OE

A1

3

(c) $f(x) = \frac{d}{dx}(F(x))$

use of

M1

$$\left. \begin{aligned} &= \frac{1}{k+1} \times \frac{d}{dx}(x+1) \\ &= \frac{1}{k+1} \quad -1 \leq x \leq k \\ &= 0 \quad \text{otherwise} \end{aligned} \right\}$$

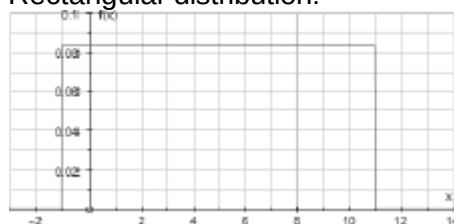
AG; $\frac{1}{k+1}$ clearly deduced

A1

2

(d) (i) $k = 11 \Rightarrow f(x) = \begin{cases} \frac{1}{12} & -1 \leq x \leq 11 \\ 0 & \text{otherwise} \end{cases}$

Rectangular distribution:



horizontal line on $[-1, 11]$

B1

$$\text{at } f = \frac{1}{12}$$

B1

2

$$(ii) \quad E(X) = \frac{1}{2}(-1 + 11) = 5$$

B1

$$\text{Var}(X) = \frac{1}{12}(11 - -1)^2 = 12$$

B1

2

$$(iii) \quad P(a_1 < X < E(X)) = P(2 < X < 5)$$

$$= (5 - 2) \times \frac{1}{12}$$

M1

$$= 0.25$$

AG

A1

2

[13]

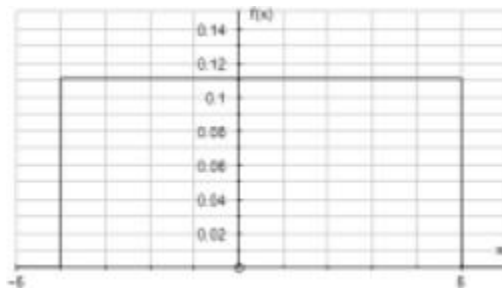
Q12.

$$(a) \quad f(x) = \begin{cases} \frac{1}{9} & -4 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

M1A1

2

(b)



horizontal line from -4 to 5

B1

for drawn at $\frac{1}{9}$

B1

2

$$\begin{aligned} \text{(c)} \quad P(X > 2) &= \frac{1}{9} \times 3 \\ &= \frac{F(5) - F(2)}{3} \\ &= 1 - \frac{2}{3} \end{aligned}$$

M1

$$= \frac{1}{3}$$

A1

2

$$\text{(d)} \quad \text{Mean} = \frac{1}{2}$$

B1

$$\begin{aligned} \text{Variance} &= \frac{1}{12} \times 81 \\ &= 6.75 \end{aligned}$$

B1

2

[8]

Q13.

(a) For a Rectangular Distribution

$$f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

$$(-0.05, 0.05) \Rightarrow$$

(explain error ± 0.05)

B1

$$\frac{1}{b-a} = \frac{1}{0.05 - (-0.05)} = \frac{1}{0.1} = 10$$

M1A1

$$(\text{Area} = 10 \times 0.1 = 1)$$

3

(b) $P(-0.01 < X < 0.02) = 0.03 \times 10 = 0.3$

M1A1

2

(c) Mean = 0

CAO

B1

Standard deviation = 0.0289

$$\frac{1}{20\sqrt{3}} \text{ OE}$$

B1

2

[7]

Q14.

(a) (i)

$$E\left(\frac{1}{X}\right) = \int_0^1 \frac{1}{x} 3x^2 \, dx = \int_0^1 3x \, dx$$

M1

$$= \left[\frac{3x^2}{2} \right]_0^1 = 1.5$$

A1

CAO

A1

3

(ii) $E\left(\frac{1}{X^2}\right) = \int_0^1 \frac{1}{x^2} 3x^2 \, dx = \int_0^1 3 \, dx$

M1

$$= [3x]_0^1 = 3.0$$

A1

$$\text{Var}\left(\frac{1}{X}\right) = 3.0 - (1.5)^2$$

dependent on previous M1

m1

$$= 0.75$$

[on their (i)] and $\text{Var} > 0$

A1ft

4

$$(b) \quad E\left(\frac{5+2X}{X}\right) = E\left(\frac{5}{X} + 2\right)$$

M1

$$= 5E\left(\frac{1}{X}\right) + 2$$

M1

$$= 5 \times 1.5 + 2$$

$$= 9.5$$

CAO

A1

$$\text{Var}\left(\frac{5+2X}{X}\right) = \text{Var}\left(\frac{5}{X} + 2\right)$$

$$= 25 \times \text{Var}\left(\frac{1}{X}\right)$$

M1

$$= 25 \times 0.75$$

$$= 18.75$$

CAO

A1

5

[12]

Q15.

$$(a) \quad E(T) = \int_0^1 t f(t) dt$$

$$= \int_0^1 4t^2 (1-t^2) dt$$

M1

$$= \left(\frac{4t^3}{3} - \frac{4t^5}{5} \right) \Big|_0^1$$

A1

$$= \frac{4}{3} - \frac{4}{5}$$

A1

$$= \frac{8}{15}$$

AG

3

(b) (i) $F(t) = P(T < t) = \int_0^t f(t) dt$

$$\int_0^t 4t(1-t^2)dt$$

M1

$$= (2t^2 - t^4) \Big|_0^t$$

$$= 2t^2 - t^4$$

A1

2

(ii) $P(\mu < T < m) = F(m) - F(\mu)$

$\Downarrow$

$$F(m) = 0.5$$

M1

B1

$$F(\mu) = F\left(\frac{8}{15}\right) = 0.4880$$

B1

$$P(\mu < T < m) = 0.5 - 0.488$$

0.5 - their $F(\mu)$

M1ft

$$\therefore = 0.012$$

A1

5

[10]

Q16.

(a) $\alpha = \frac{1}{0.4 - (-0.4)} = 1.25$

B1

1

(b) $E(R) = \frac{1}{2}(-0.4 + 0.4) = 0$

B1

$$\text{Var}(R) = \frac{1}{12}(0.8)^2 = \frac{4}{75}$$

M1

Standard deviation $= \frac{2}{5\sqrt{3}} = 0.231$

A1

3

(c) $P(|R| < 3) = P(-0.3 < R < 0.3)$

(or seen on diagram)

M1

$$= 0.6 \times 1.25$$

$$= 0.75$$

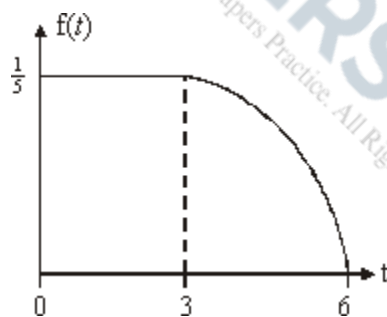
A1

2

[6]

Q17.

(a)



B1 2 axes with scales

B1 Horizontal line at 0.2 from 0 to 3

B1 Curve from 3 to 6

B3

3

(b) $P(T = 3) = 0$

B1

1

(c) $P(T \geq 3) = 1 - P(T < 3)$

M1

$$= 1 - \frac{3}{5}$$

$$= \frac{2}{5}$$

$$\int_3^6 \frac{1}{45} t(6-t) dt = \frac{2}{5}$$

A1

2

(d) $\int_0^m \frac{1}{5} dt = 0.5$

$$P(T \leq 3) = 0.6$$

M1

$$\therefore \left(\frac{t}{5}\right)_0^m = 0.5$$

$$\therefore 0 \leq \text{median} < 3$$

$$\frac{m}{5} - 0 = 0.5$$

$$\frac{1}{5}m = 0.5$$

$$m = 0.5 \times 5$$

$$m = 5 \times 0.5$$

$$m = 2.5$$

$$m = 2.5 \quad \text{AG}$$

A1

2

(e) $E(T) = \int_0^3 \frac{1}{5} t dt + \int_3^6 \frac{1}{45} t^2 (6-t) dt$

M1

$$= \left[\frac{1}{10} t^2 \right]_0^3 + \left[\frac{2}{45} t^3 - \frac{1}{180} t^4 \right]_3^6$$

A1A1

$$= \frac{9}{10} + 1.65$$

$$= 2.55$$

A1

$$\therefore P(\text{median} < T < \text{mean})$$

$$= P(2.5 < T < 2.55)$$

M1

$$= 0.05 \times \frac{1}{5}$$

$$= 0.01$$

A1

6

[14]

Q18.

(a) $P(2 \leq X \leq 3) = F(3) - F(2)$

M1

$$= \frac{27}{64} - \frac{1}{8}$$

$$= \frac{19}{64} = 0.297$$

CAO/AWRT

A1

2

(b) $f(x) = \begin{cases} \frac{3x^2}{64} & 0 \leq x \leq 4 \end{cases}$

M1A1

2

(c) (i) $E(X) = \int_0^4 x.f(x) dx$

$$= \int_0^4 \frac{3x^3}{64} dx$$

M1

$$= \left. \frac{3x^4}{256} \right|_0^4$$

A1ft

$$= 3$$

CAO

A1

3

$$(ii) \quad \text{Var}(X) = \int x^2 f(x) dx - [E(X)]^2$$

$$= \int_0^4 \frac{3x^4}{64} dx - (3)^2$$

M1

$$= \left. \frac{3x^5}{320} \right|_0^4 - 9$$

on their $f(x)$ and μ^2

A1ft

$$= 9\frac{3}{5} - 9$$

$$= \frac{3}{5} \text{ or } 0.6$$

CAO

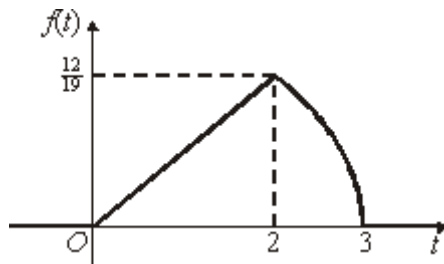
A1

3

[10]

Q19.

(a)



B1 for straight line on $[0, 2]$

B1 for curve on $[2, 3]$

B2

2

$$(b) \quad P(T < 0.5) = \frac{1}{2} \times \frac{1}{2} \times \frac{3}{19} = \frac{3}{76}$$

or by integration:

$$\int_0^{0.5} \frac{6t}{19} dt = \left[\frac{3t^2}{19} \right]_0^{0.5}$$

M1

$$= 0.0395$$

$$= \frac{3}{76} = 0.0395 \text{ (AG)}$$

A1

2

$$(c) \quad E(T) = \frac{6}{19} \int_0^2 t^2 dt + \frac{6}{19} \int_2^3 t^2 (3-t) dt$$

M1

$$= \left[\frac{2t^3}{19} \right]_0^2 + \left[\frac{6t^3}{19} - \frac{3t^4}{38} \right]_2^3$$

A1A1

$$= \frac{16}{19} + \frac{33}{38}$$

M1

$$= \frac{65}{38}$$

$$= 1.71$$

CAO

A1

5

[9]

Q20.

$$(a) \quad f(6) = \frac{2}{5} = 0.4$$

$$P(T \geq 6) = \frac{1}{2} \times \frac{3}{2} \times \frac{2}{5}$$

M1

$$= \frac{3}{10} \text{ or } 0.3$$

A1

2

(b) $\int_3^m \frac{1}{90} t^2 dt = 0.5$
Correct limits

M1

$$\int f(t) dt = 0.5$$

M1

$$\therefore \left(\frac{1}{270} t^3 \right)_3^m = 0.5$$

$$\int \frac{1}{90} t^2 dt$$

m1

$$= \frac{1}{270} t^3$$

A1

$$\left. \begin{array}{l} m^3 - 27 = 135 \\ m^3 = 162 \end{array} \right\}$$

Substitution of correct limits to obtain a cubic

m1

$$m = 5.45$$

CAO

A1f.t.

6

(c) $\int_3^6 \frac{1}{90} t^3 dt + \int_6^{7.5} \left(2t - \frac{4}{15} t^2 \right) dt$

Attempt at:

M1

(i) $tf(t)$

m1

(ii) two integrals

$$\left[\frac{t^4}{360} \right]_3^6 + \left[t^2 - \frac{4}{45}t^3 \right]_6^{7.5}$$

(iii) correct integration

A1A1

$$\begin{aligned} & (3.6 - 0.225) + (18.75 - 16.8) \\ & 3.375 - 1.95 \\ & = 5.325 \end{aligned}$$

$$\frac{13}{540} \text{ (AWRT 5.33)}$$

A1

5

[13]

Q21.

(a) $P(T = 4) = 0$

B1

1

(b) median = 3 $\Rightarrow P(T \leq 3) = 0.5$

$$P(T > 3) = \frac{1}{2} \times f(3) \times (5 - 3)$$

M1

$$\begin{aligned} &= \frac{1}{2} \times 0.5 \times 2 \\ &= 0.5 \end{aligned}$$

A1

2

(c) $P(T \geq 3.5) = \frac{1}{2} \times f(3.5) \times (5 - 3.5)$

M1

$$= \frac{1}{2} \times \frac{3}{8} \times 1.5$$

M1A1

$$= \frac{9}{32}$$

$$0.28125$$

A1

$$\therefore P(T < 3.5) = 1 - \frac{9}{32}$$

$$= \frac{23}{32}$$

0.71875 (AWRT 0.719)

A1

5

(d) $E(T) = \int_0^3 \frac{t^3}{18} dt + \frac{1}{4} \int_3^5 t(5-t) dt$

M1

$$= \left[\frac{t^4}{72} \right]_0^3 + \left[\frac{1}{4} \left\{ \frac{5t^2}{2} - \frac{t^3}{3} \right\} \right]_3^5$$

Integration attempted

M1A1

$$= \frac{9}{8} + \frac{1}{4} \left\{ \left(\frac{125}{2} - \frac{125}{3} \right) - \left(\frac{45}{2} - 9 \right) \right\}$$

Substituting limits

M1

$$= 1\frac{1}{8} + 1\frac{5}{6}$$

$$= 2\frac{23}{24}$$

2.9583 (AWRT 2.96)

A1

5

[13]

Q22.

- (a) (i) $f(t) = 2e^{-2t}$, $(t \geq 0)$
no penalty for omitting $t \geq 0$

B1

1

- (ii) Mean = 0.5
CAO

B1

Standard deviation = 0.5

CAO

B1

2

(b) $P(T > 3) = 1 - F(3)$
 $= 1 - (1 - e^{-6})$
or by integration

M1

$= 0.0248$

AWRT 0.00248

A1

2

(c) Median m satisfies $F(m) = 0.5$
 $1 - e^{-2m} = 0.5$
or by integration

M1

$e^{-2m} = 0.5$
 $-2m = \ln(0.5)$
or $e^{2m} = 2$

$m = -0.5 \ln(0.5)$
giving $0.5 \ln(2)$

A1

$(= 0.347)$

Median delay = 20.82
 $= 21$ seconds to nearest second

CAO

A1

3

[8]

Q23.

(a) Mean $\mu = \frac{a + (a + k)}{2} = \frac{2a + k}{2}$ or $a + \frac{k}{2}$

B1

Variance $\sigma^2 = \frac{((a + k) - a)^2}{12} = \frac{k^2}{12}$

CAO

B1

2

(b) Attempt at using results from (a)

M1

$$\text{Var}(X) = 3 \Rightarrow \frac{k^2}{12} = 3 \text{ or } k^2 = 36$$

or equivalent; must be stated
AG

A1

$$\therefore k = 6$$

$$E(X) = 21 \Rightarrow a + 3 = 21$$

$$\therefore a = 18$$

or equivalent; must be stated

AG

[Accept reverse arguments]

A1

3

(c) $P(20 < X < 25) = P(20 < X < 24)$
Upper limit of 24 or $(a + k)$

B1

$$= \frac{24 - 20}{k}$$

Area of rectangle of height k

M1

$$= \frac{2}{3} \text{ or } 0.67$$

CAO/AWRT

A1

3

[8]

Q24.

(a) $f(x) = \frac{1}{6}, 6 \leq x \leq 12$

$$\text{Mean, } \mu = \frac{6 + 12}{2} = 9$$

CAO

B1

$$\text{Variance, } \sigma^2 = \frac{(12 - 6)^2}{12} = 3$$

CAO

B1

2

(b) $P(X > \mu + \sigma) = P(X > 9 + \sqrt{3})$

Concept of area of a rectangle or correct integral

M1

$$\left. \begin{aligned} &= \frac{12 - (9 + \sqrt{3})}{6} \\ \text{or } &1 - \frac{(9 + \sqrt{3}) - 6}{6} \end{aligned} \right\}$$

f.t. on (a): must use $\sqrt{\quad}$ or equivalent
Wrong tail $\Rightarrow$ M1A1 f.t. A0 max
or correct integration

A1f.t.

= 0.21

AWRT; accept $\frac{3 - \sqrt{3}}{6}$ OE

A1

3

[5]

Q25.

(a) Area = 1 = $\frac{1}{2} \times 4 \times 4k$
 $\int_0^4 f(x) dx = 1$

M1

$8k = 1$

$k = \frac{1}{8}$

AG

A1

2

(b) $E(X^2) = \int_0^4 x^2 \cdot \frac{1}{8} x dx$

M1

$$= \left[\frac{1}{32} x^4 \right]_0^4$$

A1

$$= 8$$

A1

$$\text{Var}(X) = E(X^2) - E(X)^2$$

$$= 8 - \left(2 \frac{2}{3} \right)^2$$

$$= \frac{8}{9}$$

$$\left\{ \begin{array}{l} \text{ft on their } E(X^2) \\ \text{OE} \end{array} \right.$$

A1f.t.

4

$$(c) \quad \text{Var}(5X - 4) = 25 \text{Var}(X)$$

$$= 25 \times \frac{8}{9}$$

$$\left\{ \begin{array}{l} \text{for } 5^2 \times \text{their Var}(x) \\ 5^2 \times \text{Var}(x) \pm a \\ \Rightarrow \text{M0A0} \end{array} \right.$$

$$= 22 \frac{2}{9}$$

OE

A1f.t.

2

$$(d) \quad \text{Let } a = 90^{\text{th}} \text{ percentile}$$

$$\text{then } \int_0^a \frac{1}{8} x \, dx = 0.9$$

$$\left[\frac{1}{16} x^2 \right]_0^a = 0.9$$

$$\frac{1}{16} \times a^2 = 0.9$$

$$\left\{ \begin{array}{l} \text{any of these seen or} \\ \int_a^4 \frac{1}{8} x \, dx = 0.1 \end{array} \right.$$

M1

$$a^2 = 0.9 \times 16 = 14.4$$

for 14.4 seen

A1

$$a = 3.79(3\text{sf})$$

$$\begin{array}{c} 3.7947 \\ (\pm 3.79 \Rightarrow M1A1A0) \end{array}$$

A1

3

[11]

Q26.

(a) $\mu = 201$

CAO (ignore how obtained)

B1

$$\sigma^2 = 3$$

cao

B1

2

(b) $f(x) = \frac{1}{6}$

CAO; accept AWRT 0.167 (accept $h = k = \frac{1}{6}$ written or on diagram unless followed by $f(x) = 6$)

B1

1

(c) (i) $P(X = \mu) = 0$

CAO

B1

1

(ii) $P(X < 200 - \sigma) =$

$$\frac{(200 - \sigma) - 198}{6} = 1 - \frac{(204 - (200 - \sigma))}{6}$$

use of area of rectangle or integral (allow for $200 - \sigma$ or $200 - \sigma^2 < 198$ providing also $\therefore \text{prob} = 0$)

M1

$$= \frac{2 - \sqrt{3}}{6} = 1 - \frac{(4 + \sqrt{3})}{6}$$

must use their $\sqrt{\sigma^2}$ and their $f(x)$ in formula or integral

A1f.t.

$$(2 - \sqrt{3}) / 6 = 0.045$$

CAO OE / AWRT

A1

3

[7]

Q27.

(a) So that area = $(512 - 500) \times \frac{1}{12} = 1$
Area = 1 as a result

E1

or $\frac{1}{12} = \frac{1}{k - 500}$
k = 512 as a result

1

(b) Mean, $\mu = \frac{512 + 500}{2} = 506$
 CAO

B1

Variance, $\sigma^2 = \frac{(512 - 500)^2}{12} = 12$
Use of; allow even if called standard deviation

M1

Thus standard deviation, $\sigma = \sqrt{12} = 3.46$
 CAO/AWRT

A1

3

(c) (i) $P(X < 509) = \frac{509 - 500}{12}$
Attempt at area of rectangle below 509 or of integral from 500 to 509

M1

$$\frac{3}{4} = 0.75$$

CAO

M1

2

- (ii) $P(X > \mu - \sigma) = \frac{512 - (\mu - \sigma)}{12}$
Attempt at area of rectangle above
 $(\mu - \sigma)$ or of integral from $(\mu - \sigma)$ to 512
allow use of value of σ^2 only if subsequent answer is stated as 1

M1

$$\frac{512 - (506 - \sqrt{12})}{12}$$

✓ on μ and σ in formula or integral; not on σ^2

A1f.t.

$$\frac{6 + \sqrt{12}}{12} = 0.788 \text{ to } 0.789$$

AWFW

A1

3

[9]

Q28.

- (a) (i) 1
 1 cao
- (ii) 1.54
 1.54 cao
- (iii) 1.56
 1.56 cao
- (iv) 1.73
 1.73 cao
- (v) 1
 1 cao

B1

B1

B1

B1

B1

5

(b) (i) $0.5 - 0.291 = 0.209$
method

M1

0.209 cao

A1

2

(ii) $\sqrt{0.0751} = 0.2474$
method

M1

$0.274 (0.2735 \text{ to } 0.2745)$

A1

2

allow B1 for variance = 0.0751

(iii) $E(X^2) - 1.54^2 = 0.0751$
any correct equation

M1

$E(X^2) = 2.45$
 $2.45 (2.44 \text{ to } 2.45)$

A1

2

[11]

Q29.

(a) $\int_0^1 px + q \, dx$

Correct expression - ignore limits

M1

$\left[\frac{px^2}{2} + qx \right]_0^1 = 1$

Any correct integration - anywhere

M1

$\frac{p}{2} + q = 1$

Correct prrof - **ag** - allow by diagram

A1

3

(b) $E(X) = \int_0^1 px^2 + qx \, dx$
 Correct expression - ignore limits

M1

$$= \left[\frac{px^3}{3} + \frac{qx^2}{2} \right]_0^1$$

Correct method

m1

$$= \frac{p}{3} + \frac{q}{2}$$

$$\frac{p}{3} + \frac{q}{2} \text{ cao}$$

A1

3

(c) $\frac{p}{3} + \frac{q}{2} = 0.6$
 Correct equation – candidate's (b)
 (assuming $p = 1.2$ and showing $q = 0.4$ M0)

M1

$q = 1 - \frac{p}{2}$ from (a)

$$\frac{p}{3} + 0.5 - \frac{p}{4} = 0.6$$

$$\frac{p}{12} = 0.1$$

Completely correct method

m1

from (a) $q = 0.4$

$p = 1.2$ **ag** $q = 0.4$ cao

A1

3

(d) $E(X^2) = \int 1.2x^3 + 0.4x^2 \, dx$
 Correct expression for $E(X^2)$, candidate's p and
 q - allow in terms of p and q and if called variance

M1

$$= \left[\frac{1.2x^4}{4} + \frac{0.4x^3}{3} \right]_0^1$$

Correct expression for $E(X^2)$, candidate's p and q - allow in terms of p and q and if called variance

m1

$$= 0.4333$$

$$V(X) = 0.4333 - 0.6^2 = 0.07333$$

Completely correct method for σ - allow variance if called variance

m1

$$\sigma = 0.271$$

$$0.271(0.27 - 0.272)$$

A1

4

[13]

Q30.

- (a) Maximum of p.d.f.

Maximum of graph

E1

1

- (b) (i) Variance = $0.369 - 0.542^2 = 0.075236$

Method for variance

M1

$$\text{s.d.} = \sqrt{0.075236} = 0.274$$

Method for s.d. - disallow if called variance

m1

$$0.274 (0.274 - 0.275)$$

SC allow A1 for variance = $0.0752 (0.0752 - 0.0753)$

A1

3

- (ii) $1 - 0.719 = 0.281$

Use of 0.719 0.281 cao

B1B1

2

- (iii) 0.557

0.557 cao

B1

1

(iv) $0.5 - 0.483 = 0.017$

Use of 0.483, allow $1 - 0.483$

B1

Correct method

M1

0.017 cao

A1

3

[10]

Q31.

(a) $\int_{-6}^{14} k \, dy = [ky]_{-6}^{14} = k(14 - (-6)) = 20k$
 Allow by diagram – generous

M1

$20k = 1$

$k = 0.05$

0.05 ag

A1

2

(b) 4

4 cao

B1

1

(c) (i) $\int_{-6}^{14} 0.05y^2 \, dy = \left[\frac{0.05y^3}{3} \right]_{-6}^{14}$
 Any correct expression for $E(Y^2)$ – ignore limits
 Correct method – arithmetic/sign errors only

M1 m1

$= 45.733 - (-3.6) = 49.3$

49.333 (49.3 – 49.35)

A1

3

(ii) $\text{Var}(Y) = 49.3333 - 4^2 = 33.3333$

Method for Var – their $E(Y^2)$

M1

$$\text{s.d.} = \sqrt{33.3333} = 5.77$$

Requires all previous method marks in (c)
5.77(5.77 – 5.775)

m1 A1

3

(d) (i) 0.3

0.3 acf

B1

1

(ii) $\int_{-2}^2 0.05y = [0.05y^2]_{-2}^2$

Allow diagram

M1

$$= 0.2$$

0.2 ag

A1

2

(ii) 0.6

Allow diagram

0.6 acf

M1 A1

2

[14]

Examiner reports

Q3.

In attempting this question, the population of students split essentially into two parts. The major part understood the concept of $F(t)$ and its relationship to $f(t)$ and generally scored very highly. The minor part usually started by integrating $F(t)$ and thereafter gained very few marks. Amongst the better students, it was disappointing that so many failed to convert 5.79 weeks into days, as required, and equally that so many at this level differentiated t^3 and obtained $2t^2$.

Q4.

Surprisingly, this was the worst attempted question on the paper. The majority of candidates gained both marks in part (a), but many failed to recognise that the continuous random variable, T , followed a rectangular distribution. Consequently, there were many attempts using calculus to find the values and, although these were often correct, this wasted time that could perhaps have been better used in answering other more difficult questions.

Part (b) was only answered correctly by the better candidates. Many did not understand that $P(-2 < T < 2)$ was required and so simply attempted to evaluate $P(T > 2)$ with the result that 23/30 was the most prevalent incorrect answer. Too many candidates treated T as a discrete random variable with $P(T > 2) = 1 - P(T \leq 1)$ often seen, whilst others again used unnecessary integration rather than simple areas of rectangles.

Q5.

In the previous Report on the Examination, it stated that too many candidates seemed to be under the misapprehension that a request for a 'sketch' entitled them to draw a very poor quality graph or copy some unscaled diagram taken from the display on their graphics calculators. This still seemed to be the case and all such attempts lost some or all of the marks available. The axes should have been drawn with the aid of a ruler and pencil and, as a minimum, should have contained the appropriate scales; $O(0, 0)$, $(1, 0)$ and $(2, 0)$ on the x-axis and $(0, 0.5)$ and $(0, 1)$ on the y-axis. The two concave arcs were often seen as being convex or as straight lines.

In part (b), there were many correct solutions with $\int_0^1 \frac{1}{2}(x^2 + 1)dx = \frac{2}{3}$ often seen. Unfortunately, those candidates who drew straight lines in their sketches, usually ended up with $P(X \leq 1) = \frac{1}{2}(0.5 + 1) = \frac{3}{4}$; this gained no credit.

Part (c) was answered very well by many candidates with the standard of algebra and integration showing a marked improvement. However, it should be noted that, where an answer is given on the examination paper, sufficient method must be shown to ensure that full marks are awarded.

Part (d)(i) was answered very well by many candidates. However, some candidates,

having found that $\text{Var}(X) = \frac{4}{5} - \left(\frac{19}{24}\right)^2 = \frac{499}{2880}$, then were unable to write down the value

of k as 2880. Whilst most candidates obtained the correct answer in part (d)(ii), a minority

thought incorrectly that $E(X^2) = \left(\frac{19}{24}\right)^2$ even though the correct value of $\frac{4}{5}$ was given in the question. Part(d)(iii) was quite well attempted although some candidates used $\text{Var}(12X - 5) = 12 \times \text{Var}(X)$ or even $\text{Var}(12X - 5) = 12^2 \times \text{Var}(X) - 5$.

Q6.

On the whole, attempts at this question were rather disappointing. In part (a), when

candidates did demonstrate that they knew that $F\left(\frac{3c}{4}\right) - F\left(-\frac{3c}{4}\right)$ was required, the algebraic manipulation then often defeated them. There were very few candidates who realised that the given $F(x)$ was the cumulative distribution function for a *rectangular distribution*. Those who did recognise this fact were simply able to find the area of the rectangle of length $\frac{3c}{2}$ and height $\frac{1}{4c}$ to give the correct answer of $\frac{3}{8}$.

In part (b), although many candidates realised that they had to use the relationship $f(x) =$

$F'(x)$, they were then either unable to differentiate $\frac{x+c}{4c}$ with respect to x to give $\frac{1}{4c}$ and/or neglected to explain what happens when they differentiate $F(x)$ in the two regions $x < -c$ and $x > 3c$, simply stating the given answer did not suffice.

In part (c), although many candidates realised at this stage that they were dealing with a rectangular distribution and so used the appropriate formulae for $E(X)$ and $\text{Var}(X)$ from the table provided on page 11 of the booklet to obtain the required answers, many others wasted a lot of time on often fruitless attempts at integration. Also, candidates' inability to

simplify $E(X) = \frac{1}{2}(2c)$ and $\text{Var}(X) = \frac{1}{12}(4c)^2$ resulted in a loss of marks.

Q7.

In part(a), too many candidates seemed to have the wrong perception that the request for a 'sketch' entitled them to draw a very poor quality graph or copy some unscaled diagram which they had managed to take from the display on their graphics calculators. All such attempts lost some or all of the marks available. The axes and straight lines should have been drawn with the aid of a ruler and should have contained the appropriate scales: O or $(0, 0)$, $(1, 0)$ and $(3, 0)$ on the x -axis and $(0, 0.5)$ on the y -axis as a bare minimum. A straight line from $(0, 0.5)$ to $(1, 0.5)$ and a straight line from $(1, 0.5)$ to $(3, 0)$ should then have been drawn. It was often quite difficult to decide whether the sketch contained the required 'straight' lines or curves, so badly attempted were the sketches.

Similarly, part (b) was very poorly attempted with most candidates simply showing that $f(1) = 0.5$ or solving $f(x) = 0.5$ to give $x = 1$. Both of these were inappropriate and so gained no credit.

There were many good attempts at finding $E(X)$ in part (c). However when, as here, the answer is given in the question, sufficient relevant working must be shown for full credit to be obtained.

Although there were many excellent solutions to part (d), there were only a few candidates

who used the most efficient method of $1 - \frac{1}{2} \times 0.75 \times \frac{3}{16}$; most chose an alternative method which required integration. There were a large number of usually less able

candidates who wrote $P\left(X > 2\frac{1}{4}\right) = f\left(2\frac{1}{4}\right) = \frac{3}{16}$, thus gaining no marks.

Q8.

In general, the attempts at this question were very poor. It should be emphasised, yet again, that when an answer is given in a question, candidates must take extra care to explain what they are doing. This was often not seen in answers to parts (a)(ii), (a)(iii) and (b)(i), where many candidates showed insufficient working to achieve full credit.

Part (a)(i) asked candidates to sketch the graph of F . Many interpreted this as allowing the drawing of a rather scruffy freehand diagram without a scale. This was not what was intended. It was expected that scaled (approximately) and labelled axes be drawn (using a ruler and preferably a pencil) and care be taken in drawing the curved part of the sketch.

Although most candidates attempted the straight line from the origin to $(1, \frac{1}{2})$ correctly, there was much less success with the curve from $(1, \frac{1}{2})$ to $(4, 1)$. The part of the graph defined by $F(x) = 1$ was often not drawn, and the part defined by $F(x) = 0$ was rarely seen at all.

Part (a)(ii) was not explained well by many candidates. Many thought incorrectly that dividing 4 (the length of the interval over which most of them drew their sketch) by 4 would give them the value of the lower quartile, whilst others assumed that (the value of the

lower quartile) = $\frac{1}{2} \times$ the value of the median) without any explanation as to why this was the case for this distribution.

It was, however, pleasing to see good candidates state correctly that $F(q_1) = \frac{1}{2}q_1 = 0.25$,

leading to the given answer of $q_1 = \frac{1}{2}$. A variety of methods, mostly correct, were employed in part (a)(iii). The vast majority of candidates realised that the upper quartile was in the interval $[1, 4]$ and used the fact that $F(q_3) = 0.75$ to good effect. Most used different rearrangements of the resultant cubic equation, whilst others used their calculators to show that $q_3 = 1.62$ (3 sf), to achieve the required given result.

The attempts at part (b)(i) were varied and mostly disappointing, with many candidates

using the result $\alpha = \frac{1}{2}$ to show that $\beta = \frac{1}{18}$ without firstly explaining why they thought that

$\alpha = \frac{1}{2}$. Some found only one relationship between α and β (usually either $\alpha = 9\beta$ or $\alpha + 9\beta = 1$) and then simply substituted the given values of α and β to show that these values did work. This was not felt to be sufficient to gain full credit since each of these equations, taken in isolation, had infinitely many solutions, of which the given solution was just one such pair of values. However, there were a few excellent solutions, the most successful of

these first showing that on the interval $[0, 1]$, $f(x) = \alpha = F'(x) = \frac{1}{2}$. This was then followed by use of the property that f was continuous at $x = 1$ to give $f(1) = \alpha = 9\beta$, thus giving $\beta =$

$\frac{1}{18}$ as required.

Part (b)(ii) was a good source of marks, even for those candidates who gained virtually no credit on earlier parts of the question. These candidates showed good algebraic and integration skills in finding correctly that $E(X) = 1.125$. A small minority of candidates, instead of using any integration methods, used their calculators to find the correct

$$\int_0^1 \frac{1}{2} x dx + \int_1^4 \frac{1}{18} x(x-4)^2 dx$$

numerical answer, usually after stating correctly that $E(X) =$. This method gained the full 5 marks for those candidates who stated the correct numerical value but only 1 mark for those who did not. Whilst the use of calculators is encouraged in data analysis and in the evaluation of numerical expressions, candidates run the risk of losing marks if they acquire an incorrect numerical answer without any intermediate working.

Q9.

Although there was one fully correct solution seen to this question, the majority of candidates could not cope with part (c). Consequently this was the worst answered

question on the paper. In part (a), $F(0) = \frac{1}{k+1}$ was all that was required but this seemed to be beyond some candidates. In part (b)(i), candidates attempted to use the relationship

$f(x) = \frac{d}{dx}(F(x))$ in order to gain the required given answer. However, attempts at sketching the graph of f in part (b)(ii) were disappointing.

Q10.

In part (a), many candidates were able to sketch the graph of f correctly. However, 'sketch the graph' does not give candidates licence to produce rough, freehand diagrams devoid of scales. In many cases, it was very difficult to give candidates credit since it was impossible to infer whether they intended to draw straight lines or curves. Consequently, some candidates lost some or all the marks for this part of the question. In part (b)(i),

$$\int_0^2 \frac{2}{15} t dt$$

many candidates used $\frac{4}{15}$, usually gaining the correct answer of $\frac{4}{15}$. However, as the graph contained straight lines, integration, although correct, was not the most efficient method. Although there were many correct solutions seen to part (b)(ii) by a variety of correct methods, there were still some candidates who treated $f(t)$ as a discrete distribution and consequently thought incorrectly that $P(2 < T < 4) = P(T \leq 3) - P(T \leq 2)$. This method gained no credit. Some candidates, presumably from experience on past papers, first derived the cumulative distribution function, $F(t)$. Although this method gained the better candidates full marks, it was not the most efficient way of doing this part of the question. In part (c), the great majority of candidates considered the correct integrals and performed correct integration and were then able to evaluate these using correct limits to

$$2 \frac{2}{3}$$

gain the required answer of $\frac{2}{3}$. However, some candidates were unable to substitute the limits correctly in order to obtain this correct answer. Unfortunately, those candidates who thought $f(t)$ to be discrete earlier in the question continued with this incorrect premise here.

Q11.

In part (a), $F(0) = \frac{1}{k+1}$ was all that was required, but the use of $F(0) - F(-1) = \frac{1}{k+1}$ was not penalised. In part (b), most candidates realised that they needed to use the fact that $F(q_1) = 0.25$ and so went on to determine the correct expression for q_1 . Unfortunately many candidates must have thought that k was the lower quartile since they found an expression for k in terms of x . In part (c), candidates were expected to use the relationship

$f(x) = \frac{d}{dx}(F(x))$ in order to gain the required given answer. Unfortunately, attempts at

differentiating $F(x) = \frac{x+1}{k+1}$ were, in the main, unconvincing with little or no attention paid to $F(x) = 0$ and $F(x) = 1$. In part (d)(i), most candidates gained full marks. Many candidates used integration methods in part (d)(ii) which, although correctly done, did waste a lot of time, especially as the answer for $E(X) = 5$ could have been written down by simply

considering the sketch drawn in part (d)(i). Similarly, $\text{Var}(X) = \frac{1}{12}(11 - (-1))^2$ was a more efficient method of finding the variance. There were many well-explained solutions to part (d)(iii) with most candidates realising that $P(2 < X < 5)$ was required. Many candidates correctly stated that $P(X < q_1) = 0.25$ but then assumed that $P(X < E(X)) = 0.5$ without any explanation. This statement was true here since, for a rectangular distribution, the median and the mean take the same value.

Q12.

There were very many excellent solutions to this question with full marks seen on a regular basis; however some candidates seem to struggle with questions where the cumulative distribution function is given, rather than the probability density function. In part (a), some candidates were unable to differentiate the function and some treated the given function as the probability density function and attempted, usually with very little success, some sort of integration. Of the very many candidates who did manage to differentiate

$F(x)$ correctly, a few failed to quote the full answer for $f(x)$, simply giving the answer as $\frac{1}{9}$.

In part (b), those candidates who had obtained the correct answer to part (a) usually drew the correct graph. Unfortunately, some candidates showed their lack of understanding of what was given in the question by simply attempting to draw the graph of $F(x)$. Part (c) was done well by those candidates who understood the question, many using the graph that they had drawn in part (b) to help them to obtain the correct answer. For those who realised that this was a continuous rectangular distribution, part (d) was done very quickly by using the correct formulae from page 11 of the supplied booklet of formulae and statistical tables. However, some candidates attempted, usually with little success, to use integration methods in order to find the mean and variance of the given distribution.

Q13.

Part (a) caused the most problems in this question. Although many candidates were able to demonstrate why $f(x) = 10$ for $-0.5 \leq x \leq 0.5$, they were unable to indicate why the values of X had to lie in the given range. Part (b) was often done correctly, if not always by the most efficient method. For this rectangular distribution, there were far too many candidates still using integration, rather than a sketch diagram and simple geometry, to reach the required result. It was expected that candidates would use the formulae from page 11 of the booklet provided to answer part (c). Although many did, there were those who attempted integration methods with little success. Some candidates who managed to find the variance then did not go on to find the standard deviation.

Q14.

This was the least-well-answered question on the paper with many major errors of technique and understanding. It was only answered well by the more able candidates. It was a major concern that so many candidates thought incorrectly that

$E\left(\frac{1}{X}\right) = \frac{1}{E(X)}$ and $\text{Var}\left(\frac{1}{X}\right) = \frac{1}{\text{Var}(X)}$. These candidates wasted a lot of time evaluating $E(X)$ and $E(X^2)$ to no avail. There were also many examples of poor integration

techniques. When $E\left(\frac{1}{X}\right) = \int_0^1 \frac{1}{x} \times 3x^2 dx$ was correctly stated, some candidates then

$$\frac{3x^2}{3} \text{ or } \frac{3x^3/3}{x^2/2}$$

seemed unable to integrate correctly, with $\ln x \times$ often seen.

The fact that the variance cannot be less than zero seemed to have escaped the attention of some candidates who were apparently quite comfortable, not only in obtaining negative values for their variance, but also in using such values in part (b), where it was also evident that many candidates were let down by poor algebraic skills. It was expected that

candidates would first express $\frac{5+2X}{X}$ as $\frac{5}{X} + 2$ and then use

$E\left(\frac{5}{X} + 2\right) = 5E\left(\frac{1}{X}\right) + 2$ and $\text{Var}\left(\frac{5}{X} + 2\right) = 5^2 \text{Var}\left(\frac{1}{X}\right)$ to obtain answers of 9.5 and 18.75 respectively. Unfortunately many candidates could not even cope with the first step. On the other hand, there were some good attempts from evaluating

$E\left(\frac{5+2X}{X}\right)$ and $E\left(\left(\frac{5+2X}{X}\right)^2\right)$ by integration.

Q15.

Candidates seemed well able to attempt part (a) with the result that many successful outcomes were seen. Part (b)(i) was only done correctly by the most able students, with

$\int_0^1 4t(1-t^2) dt$ rarely seen otherwise. Many candidates either used 0 and 1 as their limits or simply ignored limits altogether. It was not sufficient to simply consider the indefinite integral $\int 4t(1-t^2) dt$ without consideration of the boundary conditions.

In part (b)(ii), many candidates realised that $P(\mu < T < m) = F(m) - F(\mu)$ but then went on to use the equation $2m^2 - m^4 = 0.5$ to show that $m = 0.541$ before attempting to

evaluate $\int_{\frac{8}{15}}^{0.541} f(t) dt$, instead of simply evaluating $0.5 - F\left(\frac{8}{15}\right)$.

Q16.

The use of calculus in questions of this type still seemed to be a popular, yet unnecessary, method for a small minority of candidates. Most were able to find the correct value of a in part (a), although it did defeat more candidates than expected. Part (b) was usually done well by those who used the correct formulae from the booklet. In part (c),

many candidates failed to understand the concept of the magnitude of the error.

Q17.

Statistics 2A

In general, candidates seemed to have less success with this question. In part (a), the correct graph was only seen very infrequently, with the majority of candidates drawing straight lines to represent both parts of the graph. Where a curve was drawn from $t = 3$ to $t = 6$, it was often concave rather than convex. The vast majority of candidates correctly gave the answer of 'zero' to part (b). However, a minority treated this as a discrete

distribution and used $P(X = 3) = P(X \leq 3) - P(X \leq 2) = \frac{1}{5}$. Part (c) was usually done well. However, many candidates used the more difficult integration route and, the correct answer of 0.4 was often seen. Part (d) was usually attempted quite well with most

candidates aware that the median value, m , had to satisfy $\int_0^m \frac{1}{5} dt = 0.5$. Part (e) was found difficult by all but the more able candidates. Calculating the mean to be 2.55 using the integration of two functions for the two different ranges was beyond most candidates, with several attempting to integrate one or other of the functions between 0 and 6. The final part, calculating $P(\text{median} < T < \text{mean})$ was beyond almost all but the best candidates.

Statistics 2B

In general, candidates seemed to have less success with this question than others on the paper. In part (a), the correct graph was seen very infrequently, with the majority of candidates drawing straight lines to represent both parts of the graph. Where a curve was drawn from $t = 3$ to $t = 6$, it was often concave rather than convex in nature. The great majority of candidates correctly gave the answer of 'zero' to part (b). However, a minority

treated this as a discrete distribution and used $P(X = 3) = P(X \leq 3) - P(X \leq 2) = \frac{1}{5}$. Part (c) was usually done well. However, many candidates used the more difficult integration route and, the correct answer of 0.4 was often seen. Part (d) was usually attempted quite

well with most candidates aware that the median value, m , had to satisfy $\int_0^m \frac{1}{5} dt = 0.5$. Part (e) was found difficult by all but the more able candidates. Calculating the mean to be 2.55 using the integration of two functions for the two different ranges was beyond most candidates, with several attempting to integrate one or other of the functions between 0 and 6. The final part, calculating $P(\text{median} < T < \text{mean})$ was beyond almost all but the best candidates.

Q18.

The responses to this question were very disappointing, with the great majority of candidates unable to distinguish between the distribution function, $F(x)$ as given, and the probability density function $f(x)$. Part (a) was usually only answered correctly by the most able candidates. Instead of simply evaluating $F(3) - F(2)$, the great majority integrated

$\frac{x^3}{64}$ before substituting values of $x = 3$ and $x = 2$. Part (b) was rarely attempted correctly

with differentiation of $F(x)$ to find the required form for $f(x) = \frac{3x^2}{64}$ infrequently seen. In

part (c)(i), most candidates assumed that $f(x) = \frac{x^3}{64}$, again revealing an inability to distinguish between the probability density function and the distribution function. In part (c)(ii), too many candidates only attempted to work out $\int x^2 f(x) dx$, and did not then subtract $[E(X)]^2$ in order to find $\text{Var}(X)$.

Q19.

Sketching of the graph in part (a) was not done well. The majority of candidates incorrectly

used a straight line to represent $f(t) = \frac{6}{19}t(3-t)$ for $2 \leq t \leq 3$. Of those candidates who did attempt a curve, most incorrectly drew $f(t) = kt(t-3)$. Part (b)(i) was usually done well, with integration proving to be the most popular method. Many candidates realised that the distribution of Y , the number of times that Suneil had to wait for less than 30 seconds in order to access a telephone banking service, was given by $Y \sim B(50, 0.0395)$, with the Poisson approximation, $Po(1.975)$, correctly seen to be the required distributional approximation. Unfortunately, many candidates rounded 1.975 up to 2 and then used $Po(2)$ in Table 2 to obtain their answers. This was not correct and so gained no credit. Finding the mean in part (c), when the probability density function was given over more than one range of values, gave candidates a lot of trouble. Thus only the most able candidates managed to gain full marks for this final part of the question.

Q20.

This question proved to be a good source of marks for many candidates. In part (a)(i),

most candidates used integration, with both $\int_6^{7.5} \left(2 - \frac{4}{15}t\right) dt$ and $1 - \int_3^6 \frac{1}{90}t^2 dt$ used and correctly evaluated. Only a few candidates, who had initially sketched the probability density function, used the area of a triangle in order to gain the given answer of 0.3.

There were many excellent attempts at part (a)(ii) with most candidates, realising that the

median value had to lie between 3 and 6, correctly used $\int_3^m \frac{1}{90}t^2 dt = 0.5$. Unfortunately, some candidates used zero as the lower limit and consequently gained a value of $m = 6.46$, this being outside the range of values of t for which the function $f(t) = \frac{1}{90}t^2$ is defined.

Part (b) was usually very well done with the vast majority of candidates realising that

$\int t f(t) dt$ had to be considered. Most proceeded to attempt to evaluate $\int_3^6 \frac{1}{90}t^3 dt + \int_6^{7.5} \left(2t - \frac{4}{15}t^2\right) dt$. Some candidates, unfortunately, tried to evaluate one or both of these integrals using the limits 3 and 7.5.

Q21.

In part (a) it was very disappointing to see $P(T = 4) = 0.25$ in a large number of scripts. The correct answer of 'zero' was seen very infrequently.

In part (b) there were many candidates who correctly considered $\int_0^{\infty} \frac{t^2}{18} dt = 0.5$ and usually went on to gain full marks. However there were many other unsuccessful attempts to show that the median was 3. These mainly consisted of the substitution of $t = 3$ into one of the expressions $\frac{t^2}{18}$ or $\frac{1}{4}(5-t)$, giving an answer of 0.5. This then led to a false deduction.

Many very good solutions were seen to part (c) despite the indifferent starts seen in parts (a) and (b). The most common approach was by integration methods with very few candidates using the geometry of the distribution to achieve the required result. Unfortunately, a few candidates thought that <3.5 meant that the limit to use was 3.45 or 3.49.

Many excellent attempts at the correct method to find the mean were seen in part (d). Unfortunately, some candidates could not cope with the fairly simple and straightforward integration.

Q22.

Most candidates made a good start with this question and the results for the mean and standard deviation of the exponential distribution were well known. The majority of candidates used the given formula to calculate the probability in part (b) and to find the median in part (c). A few attempted both these parts by integrating the function they had written down in part (a)(i) and this wasted time.

Q23.

In part (a), a significant number of candidates used a and k rather than a and $a + k$, or did not simplify their correct expressions. Those candidates who obtained correct simplified expressions in part (a) generally scored the three marks in part (b). Here about half the solutions involved solving $2a + k = 42$ and $k^2 = 36$ for k and a , while the others showed that $a = 18$ and $k = 6$ satisfied $E(X) = 21$ and $\text{Var}(X) = 3$. Correct answers to part (c) were the norm, with most candidates recognising that the answer equated to the rectangular area of $(24 - 20) \times \frac{1}{6}$.

Q24.

Answers to this question were often disappointing, suggesting that candidates did not understand the rectangular distribution. In part (a), some candidates treated the distribution as discrete whilst others were apparently unaware of the formulae for μ and σ^2 in the booklet. Even those candidates who answered part (a) correctly used $\mu + \sigma^2$, rather than $\mu + \sigma$, in part (b) and then stated that $P(X \geq 13) = 0$; an answer which was awarded no marks.

Q25.

Many fully correct solutions to this question were seen.

In part (a), most candidates correctly found the value of k , usually by integration. The use of the area of a triangle was surprisingly rarely seen. Since the answer was given, full credit could only be given where sufficient evidence of the correct method was seen.

In part (b), the correct form of $E(X)^2 = \int_0^4 x^2 \times \frac{1}{8} x dx$ was frequently seen. Unfortunately some candidates failed to integrate this correctly. A common incorrect response was $\frac{x^3}{3} \times \frac{1}{8} \times \frac{x^2}{2}$.

In the main, it was only the most able candidates who could cope with part (c) of this question. This part of the specification did seem to pose problems for the weaker candidates. Many candidates incorrectly assumed that $\text{Var}(5X - 4) = 25 \times \text{Var}(X) - 4$ or even $5 \times \text{Var}(X)$ instead of using $\text{Var}(aX \pm b) = a^2 \times \text{Var}(X)$.

In part (d), there were many good solutions. Most used $\int_0^a \frac{1}{8} x dx = 0.9$ to correctly calculate the 90th percentile as 3.79 correct to 3 significant figures.

Q26.

Most candidates scored the 2 marks in part (a), though a few chose to divide by 2 rather than 12 when finding the variance. It was pleasing to see the large number of correct

answers to parts (b) and (c)(i); the usual incorrect values were $\frac{1}{6}$ or $\frac{1}{7}$. In answering part (c)(ii), candidates tended to use their value of σ^2 , rather than σ and/or failed to find an actual area by not using their answer to part (b). Those candidates who sketched a diagram were often the more successful.

Q27.

Attempts at this question on a rectangular distribution were much improved over answers on previous papers. In part (a), almost all candidates reasoned correctly as to why $k = 12$

and then scored the 3 marks in part (b), though a minority used $\sigma^2 = \frac{1}{12} (b - a) = 1$ or $\sigma = 12$.

Answers to part (c) (i) were sound and, in the main, scored full marks. However, in part

(c) (ii), many candidates used $\frac{(\mu - \sigma) - 500}{12}$, often obtaining the correct value of 0.211, but then did not find the required answer of $1 - 0.211 = 0.789$.

Q28.

Although this was not a particularly high scoring question, candidates as a body seemed to prefer it to questions set on previous papers which have required integration. A few

candidates were unaware that $\int_1^2 (x - 1.54)^2 f(x) dx$ is the variance. Instead they expanded this integral and hence evaluated $E(X^2)$ for part (b)(iii). They then used this to evaluate the standard deviation in part (b)(ii).

Q29.

While the more mathematically able candidates had no problem with this question, others were unable to deal with the use of symbols p and q for constants.

Q30.

Some candidates were thrown by this question which was in an unfamiliar form. Explaining why 0.75 was the mode caused particular problems. However, the stronger candidates scored well.

Q31.

Most candidates used integration throughout this question. Apart from part (c)(i), methods based on diagrams or quoting formulae were equally acceptable. Part (d) caused the most problems, with some apparently unsure about the meaning of 'magnitude'.

